

NAME

utime, utimes – change file last access and modification times

LIBRARY

Standard C library (*libc*, *-lc*)

SYNOPSIS

```
#include <utime.h>

int utime(const char *filename,
          const struct utimbuf *_Nullable times);

#include <sys/time.h>

int utimes(const char *filename,
           const struct timeval times[_Nullable 2]);
```

DESCRIPTION

Note: modern applications may prefer to use the interfaces described in **utimensat(2)**.

The **utime()** system call changes the access and modification times of the inode specified by *filename* to the *actime* and *modtime* fields of *times* respectively. The status change time (ctime) will be set to the current time, even if the other time stamps don't actually change.

If *times* is NULL, then the access and modification times of the file are set to the current time.

Changing timestamps is permitted when: either the process has appropriate privileges, or the effective user ID equals the user ID of the file, or *times* is NULL and the process has write permission for the file.

The *utimbuf* structure is:

```
struct utimbuf {
    time_t actime;          /* access time */
    time_t modtime;         /* modification time */
};
```

The **utime()** system call allows specification of timestamps with a resolution of 1 second.

The **utimes()** system call is similar, but the *times* argument refers to an array rather than a structure. The elements of this array are *timeval* structures, which allow a precision of 1 microsecond for specifying timestamps. The *timeval* structure is:

```
struct timeval {
    long tv_sec;            /* seconds */
    long tv_usec;          /* microseconds */
};
```

times[0] specifies the new access time, and *times[1]* specifies the new modification time. If *times* is NULL, then analogously to **utime()**, the access and modification times of the file are set to the current time.

RETURN VALUE

On success, zero is returned. On error, *-1* is returned, and *errno* is set to indicate the error.

ERRORS**EACCES**

Search permission is denied for one of the directories in the path prefix of *path* (see also **path_resolution(7)**).

EACCES

times is NULL, the caller's effective user ID does not match the owner of the file, the caller does not have write access to the file, and the caller is not privileged (Linux: does not have either the **CAP_DAC_OVERRIDE** or the **CAP_FOWNER** capability).

ENOENT

filename does not exist.

EPERM

times is not NULL, the caller's effective UID does not match the owner of the file, and the caller is not privileged (Linux: does not have the **CAP_FOWNER** capability).

EROFS

path resides on a read-only filesystem.

STANDARDS

POSIX.1-2008.

HISTORY**utime()**

SVr4, POSIX.1-2001. POSIX.1-2008 marks it as obsolete.

utimes()

4.3BSD, POSIX.1-2001.

NOTES

Linux does not allow changing the timestamps on an immutable file, or setting the timestamps to something other than the current time on an append-only file.

SEE ALSO

chattr(1), **touch(1)**, **futimesat(2)**, **stat(2)**, **utimensat(2)**, **futimens(3)**, **futimes(3)**, **inode(7)**